

PENETRATION TESTING REPORT

FOOTPRINTING & NETWORK SCANNING PHASES

W2-PM-FINAL | CYBERSECURITY | NETWORKWALKS

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Program/Batch	B083E-Networkwalks
Date	26 September 2026
Modules completed	W2-PM4 (The Harvester) W2-PM5 (Zenmap Scanning)
Client/Target	1. Networkwalks (secured written permission already) 2. My own local LAN Network
Permission secured from client?	Yes
Phases covered	Phase 1: Reconnaissance & Footprinting Phase 2: Scanning & Network Discovery Phase 3-5: In Progress

1. Liability Disclaimer

I performed these activities only on systems and devices where I had secured written permission, or on devices and systems that I own myself. All the material in this report is for education and research purposes only. It should not be used to break the law. The instructor, the authors and Networkwalks are not responsible for what a reader does with this knowledge. Every action taken is the responsibility of the person taking it. Misuse of these techniques can lead to criminal charges, heavy fines, loss of employment and a permanent record. In most countries, unauthorised access is a crime even when nothing is damaged.

2. Introduction

This report covers Week 2 of my ongoing internship program at Networkwalks. In place of Module W2-PM1, I completed two footprinting tasks against the networkwalks.com domain using theHarvester tool in Kali Linux, and I also completed Module W2-PM5 by scanning my own local network with Zenmap. Together these two activities show how a security tester moves from gathering public information about a domain to mapping live hosts on a network.

The theHarvester tasks were run in Kali Linux. The first task queried only the Baidu source with a result limit of 1000, and the second task queried all available sources with a result limit of 50 per source. The Zenmap scan was run on a Windows PC against my local subnet. Every step below includes the exact command used, the result I observed and a screenshot as evidence.

3. Tools Used

The table below lists each tool used in this report and its purpose.

Tool	Purpose
Kali Linux and Windows	Operating systems used for reconnaissance and scanning activities
theHarvester	Gather emails, subdomains, hosts, IP addresses and other public information about a target domain from open sources
Zenmap (Nmap GUI)	Scan the local subnet to find live hosts, IP addresses and MAC addresses
Windows CMD (ipconfig)	Identify the local IP address, subnet mask and default gateway before scanning

4. Activities Performed

4.1 Footprinting & Reconnaissance

theHarvester is a footprinting tool that gathers information such as emails, subdomains, hosts, employee names and IP addresses from public sources including search engines, certificate logs and DNS records. It comes pre installed in Kali Linux. I used it to complete two separate tasks against the networkwalks.com domain, each with a different source configuration.

Task 1: Baidu source, limit 1000

```
theHarvester -d Networkwalks.com -b baidu -l 1000
```

The -d flag sets the target domain, -b selects the source to query, and -l sets the maximum number of results. Running this command against Baidu alone returned no IP addresses, no emails, no people and no hosts for the target domain. This shows that a single source scan can easily come back empty even for a domain that does have a public footprint, since not every source indexes every domain.

Task 2: All sources, limit 50

```
theHarvester -d Networkwalks.com -b all -l 50
```

Querying all sources produced far more activity. Several sources failed immediately because theHarvester needs an API key for them and none was configured on this machine, including bevigil, bufferoverrun, Brave Search, Censys and CriminalIP. The tool then moved on to sources that do not need a paid key, searching Subdomaincenter, Robtex, THC, Threatcrowd (which did not respond), Subdomainfinder99, Urlscan, Windvane, Yahoo, Commoncrawl and Waybackarchive.

Windvane also had no API key configured, so theHarvester fell back to a DNS based pattern search for that source and still managed to find 3 subdomains this way. By the end of the scan, theHarvester reported 3 ASNs (AS13335, AS31898 and AS46606), no LinkedIn users or links, and 4 IP addresses, including 172.67.198.228 and 192.232.216.135 which appeared in the output. No emails, people or hosts were found even with all sources enabled.

Comparing the two tasks, the all sources scan clearly gathered more information than the Baidu only scan, mainly through DNS and certificate based sources rather than through search engines. It also showed me that a real world OSINT engagement depends heavily on which API keys are available, since several sources in Task 2 could not be used at all.

4.2 Network Scanning with Zenmap

For the second activity I used Zenmap to perform network discovery on my own local network. I first identified my local IP address and subnet, then used Zenmap to find live hosts and view them on a topology diagram.

Command used

```
nmap -sn 10.0.0.2/24
```

I set the target to 10.0.0.2/24 and selected the Ping scan profile, which sends a lightweight probe to every address in the range instead of doing a full port scan. The scan covered all 256 addresses in the /24 range in about 3.47 seconds and found 2 hosts up: 10.0.0.1, which returned a MAC address of 52:55:0A:00:00:01 with an unknown vendor, and 10.0.0.2, which responded as up without a MAC address being shown, most likely because it is the machine the scan was run from.

After the scan finished, I opened the Topology tab, which drew a simple diagram with the local machine in the centre and the two discovered hosts around it, both marked with a green status indicator to show they were up.

5. Risk Analysis / Impact

Based on the information collected during the footprinting and network scanning activities, I identified the following potential risks.

#	Risk / Finding	Evidence / Observation	Potential Impact	Risk Level
1	ASN and IP address information exposed	theHarvester identified 3 ASNs (AS13335, AS31898, AS46606) and IP addresses including 172.67.198.228 and 192.232.216.135	Reveals which hosting or network providers the target uses, which can help an attacker map infrastructure	Low
2	Subdomains discoverable through DNS fallback	When the all sources scan could not query paid APIs, theHarvester still found 3 subdomains using a DNS based fallback search	Extra subdomains widen the attack surface if any of them run outdated or forgotten services	Medium
3	Baidu source returned no data	The Task 1 scan against Baidu alone found no IPs, emails, people or hosts	Low risk on its own, but shows that relying on a single source can give an incomplete picture of a target	Low
4	Several OSINT sources could not be queried	bevigil, bufferoverrun, Brave Search, Censys and CriminalIP all failed because no API key was configured	This is a limitation of the assessment rather than a finding on the target, since employees, hosts or emails may exist but were not confirmed	Low
5	Live hosts visible on local network	Zenmap identified 2 live hosts (10.0.0.1 and 10.0.0.2) on the scanned /24 subnet, one with a resolvable MAC address	Any device on the network can be discovered this way, so unknown or unauthorized devices would also be visible	Medium

The risks above are observations from the footprinting and scanning exercises, not confirmed vulnerabilities.

The practical exercises primarily involved information gathering and host discovery. No exploitation or vulnerability validation was performed as part of these two modules.

Therefore, the presence of information such as a software version, IP address or DNS record does not by itself mean that the system is vulnerable. Further authorized security testing would be required to confirm any actual vulnerability.

6. Recommendations

- **Review publicly discoverable infrastructure:** Networkwalks should periodically check what ASN, IP and hosting information about it is visible through OSINT tools like theHarvester.
- **Monitor subdomains:** The 3 subdomains found through the DNS fallback search should be reviewed to confirm they are all intended to be public and are kept up to date.
- **Use more than one OSINT source:** Since the Baidu only scan returned nothing, an assessment should not rely on a single source. Running theHarvester with all sources gives a more complete picture.
- **Configure API keys for a fuller picture:** If this exercise is repeated as part of a real engagement, keys for services such as Censys or CriminalIP would let more sources return results.
- **Perform regular internal network discovery:** The organisation should periodically scan its own network with a tool such as Zenmap to confirm which devices are active.
- **Investigate unknown devices:** Any device found during a network scan that is not recognised should be investigated and confirmed.
- **Maintain network documentation:** Network topology and device information, such as the diagram produced in this report, should be kept up to date.
- **Perform security testing with authorisation:** Footprinting and scanning should only be carried out against domains and networks where permission has already been given, as was the case here.

7. Conclusion

During Week 2 of my Cybersecurity internship, I completed practical activities covering footprinting with theHarvester and network scanning with Zenmap.

In the footprinting activity, I ran theHarvester twice against the same domain with different settings. The Baidu only search returned nothing, while the all sources search returned ASN, IP and subdomain information even though several premium sources could not be queried without an API key. This taught me that the value of an OSINT tool depends a lot on which sources and keys are actually available, and that a single empty result does not mean a domain has no public footprint.

In the network scanning activity, I used Zenmap to identify my local subnet and discover the hosts active on it. I also learned how to read a ping scan result, including MAC address and vendor information, and how to view that same result as a topology diagram.

Overall, these exercises showed me that information gathering is an important part of cybersecurity work. Before any exploitation is attempted, a tester can already learn a good amount about a target by carefully reading what public sources and simple scans reveal. I also learned that this kind of testing

must always stay inside an authorised scope, which is why both activities here were performed only against a domain with permission already secured and against my own local network.

8. Evidences Collected

Task 1: theHarvester, Baidu source, limit 1000

```
(kali@kali)-[~]
$ theHarvester -d Networkwalks.com -b baidu -l 1000
Read proxies.yaml from /etc/theHarvester/proxies.yaml
*****
*
* [theHarvester]
*
* theHarvester 4.9.2
* Coded by Christian Martorella
* Edge-Security Research
* cmartorella@edge-security.com
*
*****

[*] Target: Networkwalks.com

[*] Searching Baidu.

[*] No IPs found.

[*] No emails found.

[*] No people found.

[*] No hosts found.
```

Task 2: theHarvester, all sources, limit 50 (source queries)

```
(kali@kali)-[~]
$ theHarvester -d Networkwalks.com -b all -l 50
Read proxies.yaml from /etc/theHarvester/proxies.yaml
*****
*
* [theHarvester]
*
* theHarvester 4.9.2
* Coded by Christian Martorella
* Edge-Security Research
* cmartorella@edge-security.com
*
*****

[*] Target: Networkwalks.com

Read api-keys.yaml from /etc/theHarvester/api-keys.yaml
Failed to process bevigil search for word: 'Networkwalks.com'
Error Message:
[!] Missing API key for bevigil.
Read api-keys.yaml from /etc/theHarvester/api-keys.yaml
Failed to process bufferoverun search for word: 'Networkwalks.com'
Error Message:
[!] Missing API key for bufferoverun.
Read api-keys.yaml from /etc/theHarvester/api-keys.yaml
Read api-keys.yaml from /etc/theHarvester/api-keys.yaml
Failed to process brave search for word: 'Networkwalks.com'
Error Message:
[!] Missing API key for Brave Search.
Read api-keys.yaml from /etc/theHarvester/api-keys.yaml
Read api-keys.yaml from /etc/theHarvester/api-keys.yaml
Censys API key is missing or invalid:
[!] Missing API key for Censys ID and/or Secret.
Read api-keys.yaml from /etc/theHarvester/api-keys.yaml
Read api-keys.yaml from /etc/theHarvester/api-keys.yaml
A Missing key error occurred in criminalip:
[!] Missing API key for criminalip.
```

Task 2: theHarvester, all sources, limit 50 (results summary)

```

A error occurred while processing a "work item".
[*] Searching Subdomaincenter.
[*] Searching Robtex.
[*] Searching Thc.
No response from ThreatCrowd API for: Networkwalks.com
[*] Searching Threatcrowd.
[*] Searching Subdomainfinder99.
[*] Searching Urlscan.
[*] Windvane API key not found. Using limited unauthenticated access.
[*] API unavailable, using fallback subdomain pattern search...
[*] Found 3 subdomains using DNS fallback
[*] Searching Windvane.
[*] Searching Yahoo.
[*] Searching Commoncrawl.
[*] Searching Waybackarchive.

[*] ASNS found: 3
-----
AS13335
AS31898
AS46606

[*] No LinkedIn users found.

[*] LinkedIn Links found: 0
-----

[*] IPs found: 4
-----
172.67.198.228
192.232.216.135

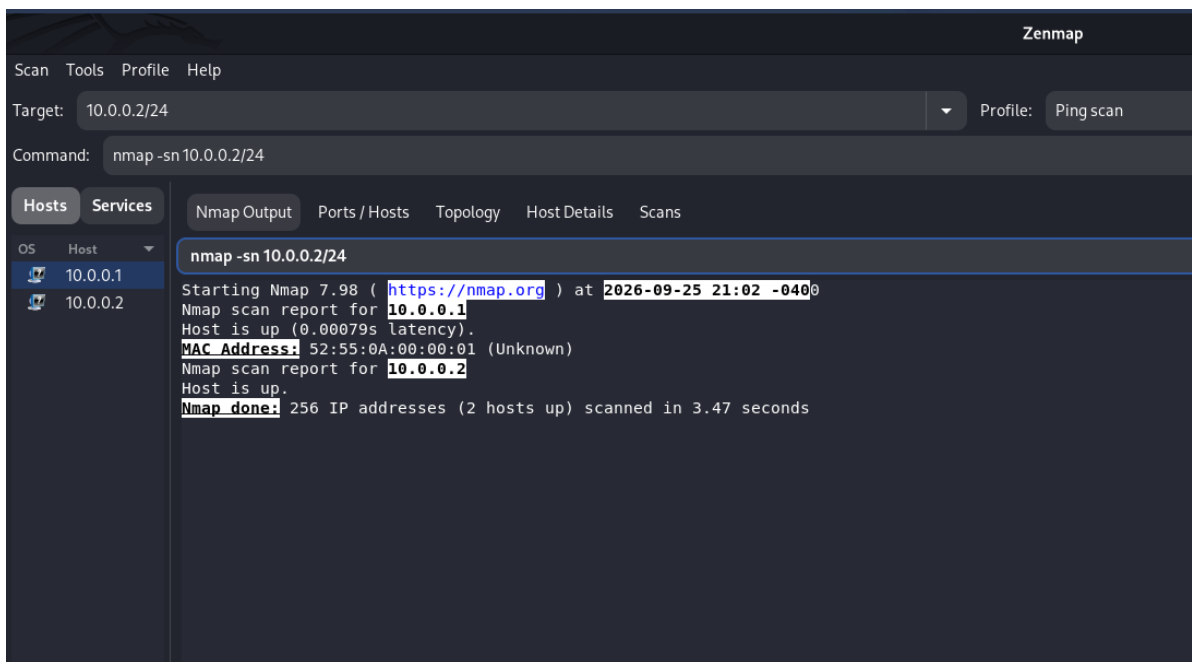
[*] No emails found.

[*] No people found.

[*] No hosts found.

```

Zenmap ping scan of 10.0.0.2/24



The screenshot displays the Zenmap application interface. At the top, there is a menu bar with 'Scan', 'Tools', 'Profile', and 'Help'. Below the menu, the 'Target' field is set to '10.0.0.2/24' and the 'Profile' is set to 'Ping scan'. The 'Command' field shows 'nmap -sn 10.0.0.2/24'. The main window is divided into two panes. The left pane shows a list of hosts under the 'Hosts' tab, with '10.0.0.1' and '10.0.0.2' listed. The right pane shows the 'Nmap Output' for the scan, which includes the following text:

```

nmap -sn 10.0.0.2/24

Starting Nmap 7.98 ( https://nmap.org ) at 2026-09-25 21:02 -0400
Nmap scan report for 10.0.0.1
Host is up (0.00079s latency).
MAC Address: 52:55:0A:00:00:01 (Unknown)
Nmap scan report for 10.0.0.2
Host is up.
Nmap done: 256 IP addresses (2 hosts up) scanned in 3.47 seconds

```

Zenmap network topology view

